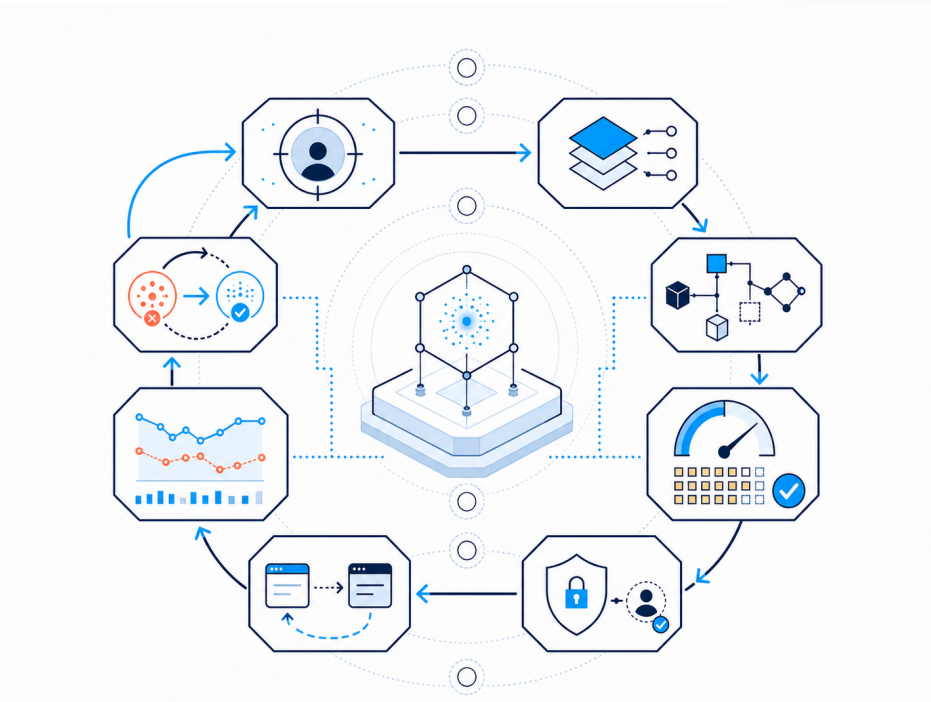
LLMOPS

Lifecycle Of LLM Apps



"It works in the demo" is the first 10%. The lifecycle is the other 90 — and **stage 8 loops back to stage 1.**

AI Coach John FOLLOW | PROITBRIDGE Strive For Better Future

REPOST

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The Demo Is 10% Of The Work

PROBLEM

The notebook that answered your question is the easy part. Everything that makes it trustworthy is the stages after it.

WHAT THE DEMO PROVES	WHAT THE LIFECYCLE ADDS
the idea is possible	is it reliable? (eval)
works on a few examples	is it safe? (guardrails)
answers the happy path	what does it cost? (monitor)
"we're nearly done"	does it stay good? (iterate)

The gap between the prototype and production is **entirely different work**, not a harder version of the same work.

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It Is A Loop, And The Loop Is The Point

CORRECTION

A pipeline runs once and stops. An LLM app lives in a loop, because the data, the model and the world keep changing under it.

→ LINE (HOW IT IS DRAWN)	🔄 LOOP (HOW IT WORKS)
scope → … → deploy → DONE	scope → … → monitor → iterate → scope
launch = the end	launch = the beginning
ships a system that can only get worse from here	prod failures become the next version

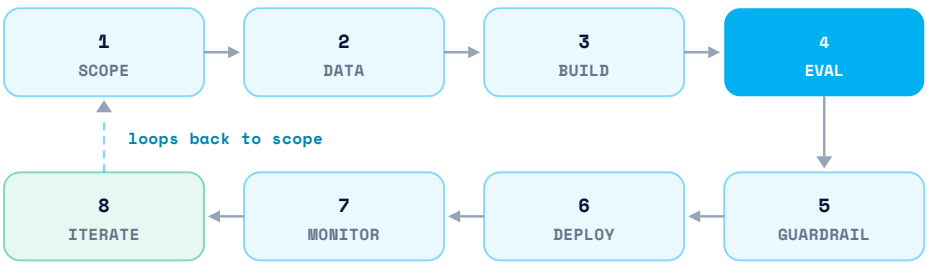
You are not building a product and finishing it. **You are building a machine that improves a product.**

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The Eight Stages

OVERVIEW

Skipping one does not remove it. It just means you handle it by accident, later.



STAGE	WHAT IT COVERS
1 scope	the use case, the user, the success metric
2 data	curate, clean, chunk and version the knowledge
3 build	prompts, RAG, model choice, orchestration
4 eval	score quality against a golden dataset, not vibes
5 guardrail	PII, injection, red-team, content filters
6 deploy	containerize, CI/CD, staging to production
7 monitor	quality, cost and drift in production
8 iterate	prod failures feed the next scope — loop to 1

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Scope: Define "Good" First

1 OF 8

A project without a success metric cannot be evaluated, and one that cannot be evaluated cannot be improved.

SCOPE DEFINES WHAT EVERY LATER STAGE MEASURES

WHO IS THE USER?	customers, not support agents
WHAT MUST IT DO?	resolve tier-one return questions
NEVER DO?	large refunds, cross-account leaks
MEASURE HOW?	resolution rate on a labelled set

THE TRAP

Starting from the tech (**"let's build a RAG chatbot"**) instead of the outcome (**"resolve 60% of return questions without a human"**). The first has no finish line.

No success metric = **nothing to evaluate = nothing to improve.**

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Data: The Variable You Own

2 OF 8

The model is a constant you rent. The data is where most RAG quality is won or lost.



GARBAGE IN, CONFIDENT GARBAGE OUT

These four decisions **directly cap the retrieval quality** you can reach in the build stage. No model choice recovers from a badly chunked, unversioned corpus.

THE TRAP

Treating data as a **one-time load**. The corpus changes, the policy updates, and a stale corpus produces confidently wrong answers.

Data is a **maintained, versioned asset**, not a file you embed once and forget.

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Build: Necessary, Not Sufficient

3 OF 8

Prompts, retrieval, model choice, orchestration. The stage everyone thinks is the whole job, and the one most over-built.

WHAT BUILD PRODUCES

prompt | retrieval | model choice | orchestration

All necessary. None of it tells you whether the answers are **right** — that is the next stage. A beautifully engineered pipeline over an unmeasured system is **a confident guess.**

THE TRAP

Polishing the build while the data is messy and the eval does not exist. **Build to a working baseline fast**, then let evaluation tell you what to improve.

Reach a baseline quickly. **Let the eval, not your intuition, drive what you improve next.**

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Eval: A Number, Not A Vibe

THE KEYSTONE

The stage that turns "it feels good" into a measurement. Without it, every change is a gamble.

CHANGE, SCORE, DECIDE

1. A change: a prompt, model or retrieval tweak.
2. Run it against the **golden set**: real inputs, labelled with good answers.
3. Score up? **Keep it.** Score down? **Revert it.**

THE TRAP

Trusting a demo. A system that looks great on **5 examples** can be **60% accurate on 200**, and you will never know without measuring.

The golden set is **the keystone**. It gates every deploy and unlocks monitoring and the flywheel.

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Guardrail: Safe Is Not The Same As Working

5 OF 8

A correct answer that leaks another customer's data is still a breach. Safety is its own stage.

CONCERN	THE GUARDRAIL
PII	never emit another customer's data
prompt injection	retrieved text is data, not commands
irreversible actions	**human approval on large refunds**
harmful content	filter input and output

A helpful model is **not automatically a safe one.** An unguarded agent is one crafted email away from a breach.

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Deploy: Ship Plus The Gates

6 OF 8

Getting it live is solved. Getting it live with an eval gate, secrets handled and a rollback is the actual job.



SHIP PLUS THE GATES THAT MAKE IT SAFE

The eval gate fails the pipeline if quality dropped. Rollback is the **previous revision, one command.**

THE TRAP

Deploying like a normal web app: straight to prod, no eval gate, secrets in the container. AI's failure mode is a **silently worse model, not a crash**, and only the eval gate catches it.

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Monitor: Watch Quality, Not Just Uptime

7 OF 8

The service can be perfectly up and quietly wrong. Only quality monitoring sees it.

INFRA (NECESSARY)	QUALITY (THE AI LAYER)
is it up?	are the answers still good?
latency, error rate	sample + score prod traffic
total cost	drift, cost per resolution

Up and fast and **wrong** is invisible to infra monitoring alone. Watch the answers, not just the servers.

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Iterate: The Flywheel

THE LOOP CLOSES

Every production failure is free training data for the next version. The teams that win close this loop fastest.

THE DATA FLYWHEEL

1. **Monitor finds a failure** — the question it got wrong.
2. Add it to the golden eval → **it can never regress again.**
3. Fix the prompt, the data or the retrieval.
4. Deploy, monitor, find the next failure → **loops forever.**

The product improves **from its own mistakes.** That is the moat a static competitor cannot copy.

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Three Things In Every Stage

CROSS-CUTTING

Cost, governance and versioning are not stages. Forgetting them is a whole-lifecycle failure.

COST

Every stage has a lever: storage, experiments, eval runs, inference, monitoring. Ignore it and you ship a product that is **economically dead.**

GOVERNANCE

Who accesses what data, what the model may do, how decisions are logged, what compliance applies. **A constraint on every stage.**

VERSIONING

Prompts, data, model and eval sets all change. When quality drops, you need to know **exactly what changed.**

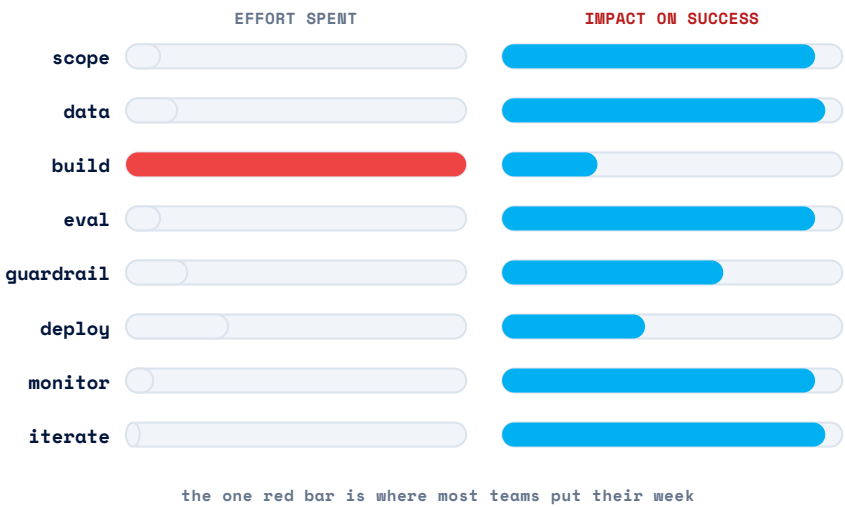
Cheapest built in from stage one, costliest retrofitted after launch.

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Where Effort Goes vs Where Impact Lives

MISMATCH

Effort goes to the visible, satisfying stage. Success is decided by the invisible ones.



the one red bar is where most teams put their week

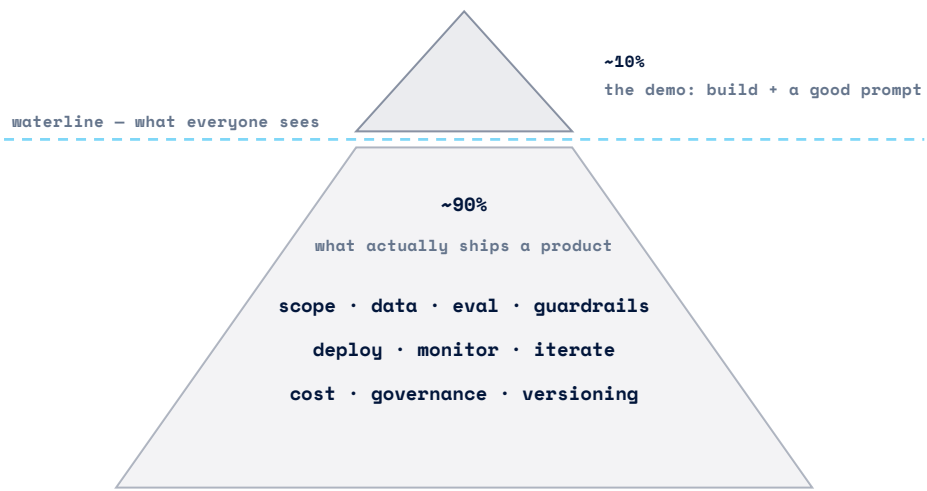
↔ Effort clusters on **BUILD.** Impact lives in **DATA, EVAL, MONITOR, ITERATE.** Reallocate.

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The Demo-To-Production Iceberg

THE GAP

From "works in the notebook" to "works for every user, safely, affordably, next month" is most of the project.

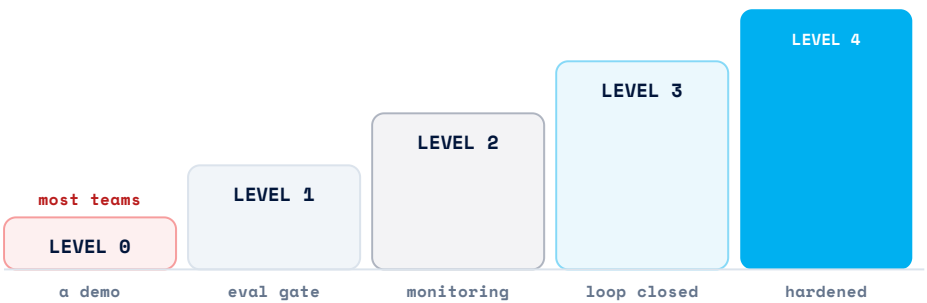


The demo proves it **can** work. The 90% makes it **keep** working.

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Where Is Your Team On The Lifecycle? MATURITY

Most teams are further left than they think. You cannot skip levels.



0 **A demo, shipped.** No eval, no monitoring. "It worked when we tried it."

1 **+ an eval gate before deploy.** "We know if a change made it worse."

2 **+ production quality monitoring.** "We know when it degrades in the wild."

3 **+ the iterate loop closed.** "Failures become eval cases automatically."

4 **+ cost, governance, versioning built in.** "Sustainable, auditable, reproducible."

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Where To Start At Level 0

GETTING STARTED

You do not fix the whole lifecycle at once. You add the one stage that unlocks the rest.

THREE QUESTIONS, IN ORDER

Q1 **Have an eval set (labelled inputs + good answers)?** No → **build it first.** 20–50 real cases. Nothing works without a way to measure. **(Level 1)**

Q2 **Monitor answer quality in production, not just uptime?** No → add it. Sample prod traffic, score it against the same eval. **(Level 2)**

Q3 **Do prod failures automatically become new eval cases?** No → close the loop, the flywheel. **(Level 3)** Yes → harden cost, governance, versioning. **(Level 4)**

The **eval set is the keystone.** It unlocks deploy gates, monitoring and the flywheel. Build it before anything else.

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Symptom To Stage

DEBUG TABLE

What you observe, and which stage of the lifecycle is the real cause.

SYMPTOM	CAUSE	STAGE
Quality varies, no one knows why	no evaluation	eval
A change made it worse, unnoticed	no eval gate on deploy	deploy
Great in testing, poor in prod	no production monitoring	monitor
Answers stale after a policy edit	corpus not versioned	data
Same mistake keeps happening	failures not fed back	iterate
Correct answer leaked an account	safety not its own stage	guardrail
Endless building, never shipping	no success metric	scope
Works but costs too much	cost ignored across stages	cross-cut

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Nine Ways To Break The Lifecycle

ANTI-PATTERNS

Almost every one is "treat the loop as a line."

ANTI-PATTERN	FIX
Ship the demo, call it done	**treat launch as stage 6 of 8**
Build first, define success later	scope and the metric come first
Evaluate by vibes	a golden set, scored every change
Safety as a final polish	guardrails are their own stage
Deploy straight to prod	eval gate + staging + rollback
Monitor uptime only	monitor answer quality too
Fix bugs, do not capture them	every failure becomes an eval case
Data loaded once, never maintained	version and update the corpus
Cost/governance as afterthoughts	build them in from stage one

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Build The Loop, Not The Demo

CHEATSHEET

An LLM app is not a thing you finish. It is a loop you operate.

STAGE	THE ONE LINE THAT MATTERS
1 scope	define "good" as a number before you build
2 data	a versioned asset, not a one-time load
3 build	reach a baseline fast, then let eval guide you
4 eval	a golden set gates every change; the keystone
5 guardrail	safety is its own stage, not a final polish
6 deploy	eval gate + staging + one-command rollback
7 monitor	watch answer quality, not just uptime
8 iterate	failures become eval cases; loop to stage 1
cross-cutting	cost, governance, versioning through all eight

The loop is the product. The demo was never the point.